ISTA 322

Final Project: Transforming Spotify Data

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Table of Contents

01 Introduction

02 Spotify API

03 Amazon S3

04 Billions Club

05 SQL Database

06 Queries

Introduction

Motivation

In honor of the 2024 Spotify Wrapped, this project explores my top artist of the year and all-time favorite artist — BTS! I'll explore their present-day popularity as mainstream artists along with their most-streamed songs on Spotify.

Goal

The end goal will be to determine the songs that have made it to the "Billions Club" and songs that are coming close to achieving this milestone, including an estimate of how much longer they'll take to do so!

Spotify API JSON Data

Extract Spotify data

- Access artist Spotify URI with the Spotipy Python library
- Extract artist info - artist name, artist ID, genre, followers, popularity
- Extract track info - track name, track ID

Normalize JSON data

- Turn JSON data into a dataframe
- Normalize ‘artists’ path further to get nested artist info
- Keep top track info while showing info on BTS and various collab artists

Clean & merge data

- Drop unnecessary columns
- Filter out collab artists so collab songs don’t appear more than once
- Merge artist info dataframe with track info dataframe

top_BTS_tracks

	artist_name	artist_id	genre	followers	popularity	track_name	track_id
0	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	Dynamite	5QDLhrAOJJdNAmCTJ8xMyW
1	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	My Universe	46HNZY1i7O6jwTA7Slo2PI
2	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	Left and Right	5Odq8ohlgIbQKMZivbWkEo
3	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	FAKE LOVE	6m1TWFMeon7ai9XLOzdbiR
4	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	Butter	6jjYDGxVJsWS0a5wIVF5vS
5	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77096667	87	Boy With Luv	4a9tbd947vo9K8Vti9JwcI

Amazon S3 CSV Data

Download HTML

- Collect dataset from BTS - Spotify Top Songs
- Data extracted on 12/3, so might not reflect real-time streams
- Download in HTML format and convert it to a CSV file

Extract CSV data

- Upload CSV file to Amazon S3 and make access public!
- Extract CSV data using Pandas Python library
- Raw data frame has misplaced column names and extra rows

Clean & merge data

- Drop unnecessary columns and rows
- Replace columns with correct headers and rename columns
- Modify track names in table to match official Spotify titles to avoid merging issues

BTS_track_streams

	track_name	total_streams	daily_streams
0	Dynamite	1,969,980,461	453, 304
1	My Universe	1,366,729,066	390,873
2	Butter	1,309,351,690	250,357
3	Boy With Luv	1,214,241,450	223,789
4	Left and Right	1,032,798,321	357,065
5	FAKE LOVE	855,840,480	282,977

BTS_songs

	artist_name	artist_id	genre	followers	popularity	track_name	track_id	total_streams	daily_streams
0	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	Dynamite	5QDLhrAOJJdNAmCTJ8xMyW	1,969,980,461	453, 304
1	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	My Universe	46HNZY1i7O6jwTA7Slo2PI	1,366,729,066	390,873
2	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	Butter	6jjYDGxVJsWS0a5wIVF5vS	1,309,351,690	250,357
3	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	Boy With Luv	4a9tbd947vo9K8Vti9Jwcl	1,214,241,450	223,789
4	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	Left and Right	5Odq8ohlgIbQKMZivbWkEo	1,032,798,321	357,065
5	BTS	3NrfpeOtUJi4K4DXYWgMUX	k-pop	77,096,667	87	FAKE LOVE	6m1TWFMeon7ai9XLOzdbiR	855,840,480	282,977

Road to the Billions Club

Make subtable

- Make copy of final dataframe
- Include only track column and streams columns
- Filter out top songs that have already reached 1B streams

Calculate time to 1B

- Days to 1B = remaining streams / daily streams
- Months to 1B = number of days / 30
- Years to 1B = number of days / 365

Clean up dataframe

- Drop unnecessary columns
- Round calculations to nearest whole integer
- Add commas back to # of streams columns for ease of reading

road_to_billions

	track_name	total_streams	daily_streams	days_to_1B	months_to_1B	years_to_1B
5	FAKE LOVE	855,840,480	282,977	509	17	1
6	Life Goes On	758,382,023	199,279	1212	40	3
7	Permission to Dance	707,136,995	156,362	1873	62	5
8	Spring Day	532,723,829	169,532	2756	92	8
9	Run BTS	481,349,650	179,510	2889	96	8

SQL Relational Database

Import SQL functions

- Get connection info using AWS endpoint for MySQL server
- Define functions using MySQL Python library
- Get connection and cursor, query data, table head info, etc.

Make empty tables

- Write SQL query to make table for each dataframe
- Get conn and cur, execute query, and commit table
- Check that all tables and column names are in the database

Load data into tables

- Convert each row in dataframe to a tuple using Numpy library
- Insert tuples and upload data into each empty table
- Check that data is in the tables and run test queries on them

SQL Relational Database

BTS_SONGS
ARTIST_NAME
ARTIST_ID
GENRE
FOLLOWERS
POPULARITY
TRACK_NAME
TRACK_ID
TOTAL_STREAMS
DAILY_STREAMS

ROAD_TO_BILLIONS
TRACK_NAME
TOTAL_STREAMS
DAILY_STREAMS
DAYS_TO_1B
MONTHS_TO_1B
YEARS_TO_1B

SQL Queries

- Most streamed song overall:

```
""" SELECT track_name, total_streams
FROM bts_songs
ORDER BY total_streams DESC
LIMIT 1; """
```

- Average daily streams for all top songs:

```
""" SELECT AVG(daily_streams) AS average_daily_streams
FROM bts_songs; """
```

- Number of top songs below 1B streams:

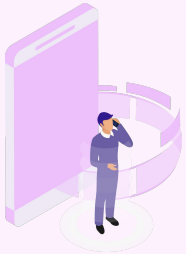
```
""" SELECT COUNT(track_name) AS tracks_below_1B
FROM road_to_billions; """
```

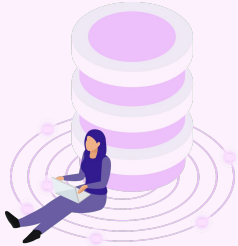
- Top song with shortest time to 1B streams:

```
""" SELECT track_name, daily_streams, days_to_1B
FROM road_to_billions
ORDER BY days_to_1B ASC
LIMIT 1; """
```

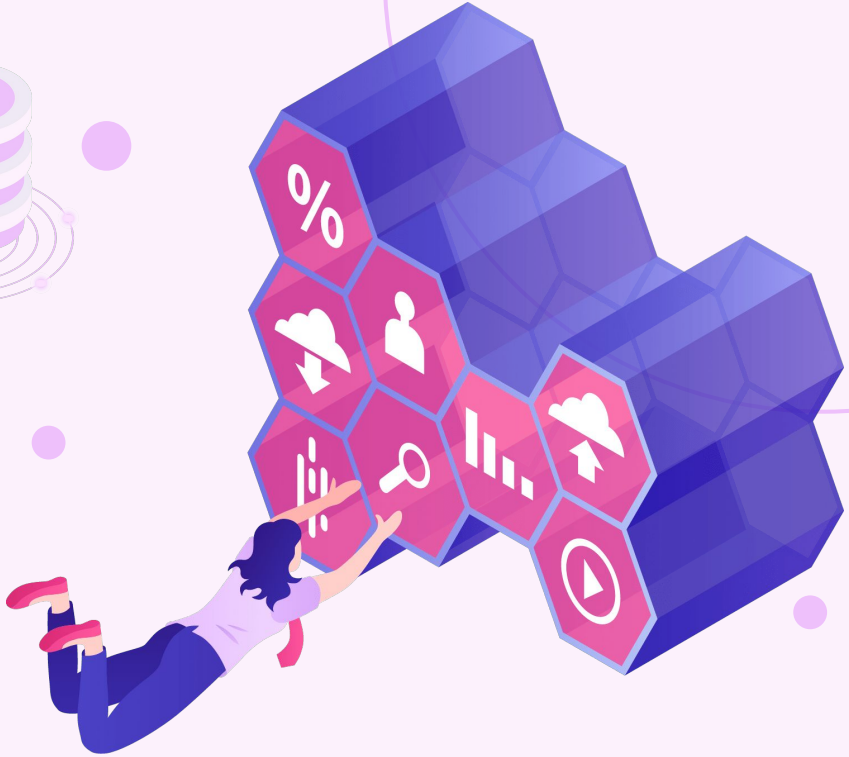
- Songs that will take over 5 years to reach 1B:

```
""" SELECT track_name, total_streams, years_to_1B
FROM road_to_billions
WHERE years_to_1B > 5; """
```





Thanks!



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